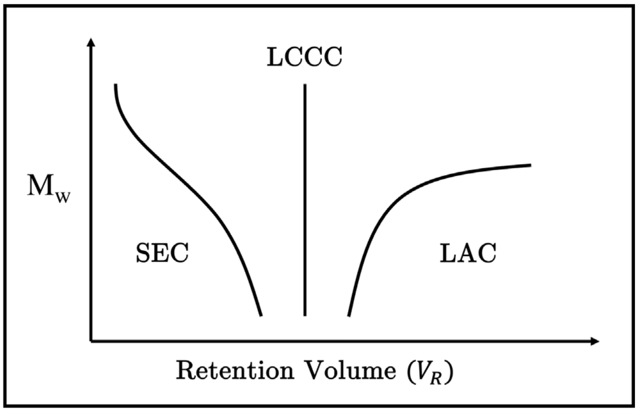

Fig. 1. Illustration of the three elution modes in polymer liquid chromatography: Size Exclusion Chromatography (SEC), Liquid Adsorption Chromatography (LAC), and Liquid Chromatography at the Critical condition (LCCC).

chemical or physical structure rather than M_w, making LCCC indispensable for characterizing complex polymer systems. LCCC is particularly valuable for analyzing a wide range of polymer characteristics that are challenging to distinguish using other chromatographic methods.

Numerous dedicated researchers have explored and identified critical conditions for polymers relevant to their fields. However, these investigations are dispersed across a wide range of chromatographic journals, resulting in a fragmented and scattered body of literature. Since its introduction in the late 1980s, LCCC has been referred to by various names and initialisms, further contributing to the lack of cohesion in the field: LC-CC, polymer critical condition (polymer CC), critical polymer chromatography (CPC), critical solvent condition (CSC), critical adsorption point (CAP, CPA, LC CAP), liquid adsorption critical chromatography (LACC), liquid chromatography at critical conditions of adsorption (LCCCA, LACCC), and liquid chromatography of polymers at the exclusion-adsorption transition point (LC EATP, LC PEAT) [1]. Though each name emphasizes a specific aspect of LCCC, they nevertheless widen the search terms necessary for research.

Being that each addition to the field is reported individually, as the field grows, the depth at which a researcher must search to find specific conditions also grows. Currently, a researcher would have to begin with an extensive literature review to find the critical conditions relevant or similar to their polymer of choice. However, the scattered nature of existing LCCC data poses significant challenges to researchers seeking to adopt or optimize this technique. No unifying resource exists to catalog and report all LCCC findings that are publicly available, and such absences are being felt throughout the polymer field [3]. Without a unifying resource, researchers would have to work with fragmented information, making the process of finding LCCC conditions a far more daunting task.

Additionally, the critical condition identified in LCCC contains rich physiochemical information about polymers, solvents, and stationary phases. Since this field is fundamentally data-centric, a collection of LCCC data can be further mined to reveal the underlying principles/properties that may not be easily observed. Without a dedicated online platform to aggregate these conditions, the integration of machine learning into LCCC conditions would remain severely limited by the lack of accessible and comprehensive data.

To address this challenge and others that surround it, we have developed a centralized, dynamic online platform with a comprehensive database of polymer liquid critical chromatography critical conditions titled *PolyCrit*. This website organizes and currently reports 428 critical conditions in its initial release; offers a user-friendly, searchable interface for ease-of-access; and welcomes current researchers to add to the database to increase the breadth of data it contains.

2. Background

Liquid chromatography encompasses a wide range of methods, including Size Exclusion Chromatography (SEC) and Liquid Adsorptive Chromatography (LAC), each distinguished by its separation mechanism. SEC separates molecules primarily based on size, utilizing a porous stationary phase where smaller molecules are retained longer due to their ability to enter the pores. This method is particularly effective for determining M_w distributions of polymers and proteins. In contrast, LAC separates molecules based on chemical interactions with the stationary phase, such as hydrophobic, polar, or ionic interactions. This makes LAC well-suited for analyzing chemical properties and structural nuances in complex mixtures.

While SEC and LAC are widely used, their limitations become evident when characterizing polymers. Polymers, due to their high range of M_w's and propensities for chemical interaction, often require hybrid or alternative approaches for effective separation. Two-dimensional liquid chromatography (2D-LC) and solvent/temperature-gradient interaction chromatography (S/TGIC) are examples of advanced methods that address these challenges [4].

However, LCCC offers a specialized solution by enabling polymer separations independent of M_w under critical conditions. This capability allows researchers to distinguish between polymers with similar M_w but differing structures. For instance, LCCC can separate linear polymers from branched or cross-linked polymers, even when these species have similar M_w. LCCC's sensitivity to structural differences extends its utility to analyzing branching degrees and other architectural features of polymers. Moreover, removing one of the most significant confounding variables in separations, M_w, provides researchers with a practical, hands-on approach to prepare highly pure polymer solutions for secondary separations, eliminating the need for industrial equipment or manufacturing controls [5].

The theory of LCCC relies on achieving a "critical condition" where the elution of a polymer is unaffected by its M_w but depends on its chemical interaction with the stationary phase of column substrates in chromatographic system. A full elaboration of the theory behind LCCC can be found in a previous work of ours [6], but we give a brief summary of the necessary terminology and equations here.

The Critical Adsorption Point (CAP) in liquid chromatography represents a pivotal transition in how polymer chains interact with a solid surface or a stationary phase. When polymer chains approach a surface, their behavior depends on a balance between entropic and enthalpic effects. In the absence of attractive interactions (enthalpic gains), the chains lose configurational entropy as they are restricted near the surface, leading to what is known as a depletion layer, where polymer density near the surface is lower than in the bulk phase.

As the strength of polymer-surface interaction increases, there comes a critical interaction strength where the entropy loss is exactly balanced by the enthalpic gain. At this point, the polymer density profile near the surface becomes flat, meaning there is neither depletion nor excess density. This critical interaction strength defines the CAP. Beyond the CAP, when the enthalpic gain outweighs the entropy loss, polymer chains begin to adsorb preferentially onto the surface, leading to an excess density layer, where the concentration of polymers near the surface exceeds that in the bulk.

The retention volume V_R of polymers in chromatography is related to the partition coefficient K by $V_R = V_0 + KV_P$, where V_0 is the void (or mobile phase) volume and V_P is the stationary phase's pore volume, respectively [7]. Using the Gaussian chain model, one can approximate the value for K [4],

$$K = 1 + \frac{2R}{D}\left[\frac{\exp\left((cR)^2\right)[1-\mathrm{erf}(-cR)]-1}{cR} - \frac{2}{\sqrt{\pi}}\right] \quad (1)$$

where $\mathrm{erf}(x)$ is the error function,

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x \exp\left(-t^2\right) dt. \quad (2)$$